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The 2023 Orthopaedic Research Society's International Consensus Meeting on musculoskeletal infection: Summary from the host immunity section

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Abstract

Musculoskeletal infections (MSKI), which are a major problem in orthopedics, occur when the pathogen eludes or overwhelms the host immune system. While effective vaccines and immunotherapies to prevent and treat MSKI should be possible, fundamental knowledge gaps in our understanding of protective, nonprotective, and pathogenic host immunity are prohibitive. We also lack critical knowledge of how host immunity is affected by the microbiome, implants, prior infection, nutrition, antibiotics, and concomitant therapies, autoimmunity, and other comorbidities. To define our current knowledge of these critical topics, a Host Immunity Section of the 2023 Orthopaedic Research Society MSKI International Consensus Meeting (ICM) proposed 78 questions. Systematic reviews were performed on 15 of these questions, upon which recommendations with level of evidence were voted on by the 72 ICM delegates, and another 12 questions were voted on with a recommendation of "Unknown" without systematic reviews. Two questions were transferred to another ICM Section, and the other 45 were tabled for future consideration due to limitations of available human resources. Here we report the results of the voting with internet access to the questions, recommendations, and rationale from the systematic reviews. Eighteen questions received a consensus vote of ≥90%, while nine recommendations failed to achieve this threshold. Commentary on why consensus was not achieved on these questions and potential ways forward are provided to stimulate specific funding mechanisms and research on these critical MSKI host defense questions.

For affiliations refer to page 528.

KEYWORDS
host immunity, International Consensus Meeting, musculoskeletal infection (MSKI), osteomyelitis, periprosthetic joint infection, surgical site infection

1 | INTRODUCTION AND BACKGROUND

Musculoskeletal infections (MSKI) continue to be the bane of orthopedic surgery, as their poor clinical outcomes have great costs to patients and healthcare systems. Most troubling is our inability to reduce the approximately 1% infection rate following elective surgery,[1–4] even with the most rigorous prophylaxis and aseptic surgical systems (e.g., outcomes from the Surgical Care Improvement Project (SCIP)[2]), suggesting that this may be due to host factors. Moreover, reinfection rates and persistent infection following revision surgery are still as high as 33%,[3,5,6] which also suggests that these patients do not have a protective adaptive immune response. It is also noteworthy that the major pathogens responsible for MSKI are bacteria (*Staphylococcus aureus*, *Streptococcus agalactiae*, and *Pseudomonas aeruginosa*, etc.[7]) that cause serious infectious diseases in other organ systems, and that all efforts to develop effective human vaccines have failed.[8] Thus, major knowledge gaps in our fundamental understanding of microbial pathogenesis and the host response to virulence must be filled if we are to develop effective interventions for MSKI. Additionally, it has been acknowledged that a major cause of human vaccine failure is the lack of preclinical model systems that faithfully replicate immune responses in real-world clinical situations.[9,10] Therefore, technical and conceptual innovations are needed.

Given that elucidation of all critical knowledge of MSKI cannot be obtained from randomized controlled clinical trials (Level 1 evidence) due to lack of feasibility, overwhelming costs, and ethical concerns, the orthopedic community turned to International Consensus Meetings (ICM) to obtain the most current information and establish priorities from expert opinions with its first ICM focused on prosthetic joint infection (PJI) in 2013.[11] This meeting was a tremendous success by several criteria including: (i) establishing an initial precedent with a published description of the consensus process;[12] (ii) changing clinical practice with the first definition of PJI that was formalized in 2018;[13] and (iii) establishing a body mass index > 40 as a risk factor for PJI.[14] Moreover, this led to a second ICM 5 years later that covered all aspects of MSKI (ICM 2018, https://icmphilly.com). Importantly, two research workgroups containing Orthopaedic Research Society (ORS) members produced publications on the General Assembly questions,[14] and on the Biofilm questions.[15] There was also a third MSKI ICM in 2019 focused on antibiotic-loaded bone cement.[16]

These initial ICM produced a new international community of investigators that launched the ORS MSKI research interest group (RIG) at the 2020 Annual Meeting in Austin, TX. At this time, the idea of holding an ORS MSKI ICM focused on preclinical questions was born, as animal modeling had largely been neglected in the prior MSKI ICM, and the painful history of vaccine development without careful attention to face, construct, and predictive value of the preclinical models used was paramount.[17,18] Another strong motivation for this topic came from the ICM 2018 results, which determined that of the General Assembly topics (microbiology/antimicrobials, biomaterials/implants, surgery/clinical care, and immunology/host immunity), "our knowledge of host immunity is by far the most limited, and it is the only discipline that has not made a major contribution to impact MSKI patients."[14] This was also reflected in the ICM 2018 voting, in which only 62% of delegates agreed that immunotherapy and immunoprophylaxis could be used to prevent biofilm formation and implant-associated infections (https://www.ors.org/wp-content/uploads/2019/01/Question-5.pdf).

In recognition of the importance of MSKI immunology, the ICM created a Host Immunity Section that originally drafted 78 questions. Delegates volunteered to perform systematic reviews on up to three of these questions. In the end, systematic reviews were performed on only 15 of these questions, upon which recommendations with the level of evidence were voted on by the 72 ICM delegates. Another 12 questions were voted on with a recommendation of "Unknown" without systematic reviews, and the remainder were either transferred to another ICM Section or tabled due to the lack of human resources for performing systematic reviews. Here we report the results of the voting with internet access to the questions, recommendations, and rationale from the systematic reviews. Eighteen out of 27 questions received a consensus vote of ≥90%, while the remaining nine recommendations failed to achieve this high agreement threshold consensus vote, which was arbitrarily set by the authors to manage the space limitations of *JOR* Consensus Articles. Commentary on why consensus was not achieved and potential ways forward are provided to stimulate specific funding mechanisms and research on these critical MSKI host defense questions.

2 | METHODOLOGY

The main objective of the "Host Immunity" Section of the 2023 ORS MSKI ICM was to bring together microbiologists, immunologists, clinicians, and FDA regulators from around the world to assimilate the best available data on preclinical research of host–pathogen interactions during orthopedic infections. The first step, led by the Section Chair, Dr. Edward M. Schwarz, was to identify and recruit host immunity experts from the ORS RIG and beyond. These experts agreed to serve as delegates tasked to identify the controversies and challenges related to topical preclinical research, then seek consensus on those issues using the Delphi methodology.[9] This has proven to be a critical tool by which thought leaders debate all existing knowledge to derive "general agreement" in response to clinical care-driven questions.

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An initial pool of 17 ORS MSKI RIG members and 11 distinguished non-ORS members (Table 1) generated 78 questions addressing preclinical research related to MSKI host immunity. Over 24 months, 22 delegates (the authors of this paper) volunteered to perform a systematic literature review, data summary, and an independent narrative response written from the perspective and experience of each authoring delegate on 15 of these questions. Experts also deemed that there was no peer-reviewed literature on 12 questions and recommended that the questions should be voted on as "Rapid Fire" with a recommendation of "Unknown" without systematic reviews. Four questions were eliminated due to redundancy with other questions. Two questions, "Which cytokines need to be studied in animal models of MSKI?" and "What is known about sex and animal strain differences in animal models of MSKI?" were transferred to the "Animal Models" Section of the ICM. The remaining 45 questions (Table 2) were not addressed due to limited human resources for systematic reviews.

The responses to the 15 questions were reviewed by a facilitator and combined into a single document that was sent via email to 119 potential delegates who indicated that they would be interested in the ICM voting. An authoring delegate then presented the question with the response and strength of recommendation to the ICM in person and via live internet streaming during the 2023 ORS MSKI RIG meeting in Dallas, TX, on February 10, 2023. Unfortunately, information technology issues and confusion over wording and grammar invalidated the live voting, and the ICM decided to redraft several questions so that they could be clearly answered with an "Agree," "Disagree," or "Abstain" vote. Delegates were advised that an "Agreement" vote means complete agreement with the recommendation and the level of evidence cited, and that a "Disagree" vote means disagreement with the recommendation and/or the level of evidence cited. In the end, 72 ICM delegates voted on all the questions, and the results for each question were tabulated by the number and % who agreed, disagreed, or abstained.

The ICM set an arbitrary threshold of ≥90% agreement to mean that ICM consensus on the question and recommendation have been achieved and that further discussion without new information is not warranted. The ICM also decided that authors of the recommendations that did not achieve ≥90% agreement should share their thoughts on what the potential controversies are and provide insights on the way forward as outlined below.

3 | RESULTS

The "Host Immunity" questions that included systematic reviews (Host Immunity) and those without systematic reviews (Rapid Fire) with the ICM's voting results are summarized in Table 3. A complete list of all 65 ICM questions with recommendations and rationale (https://ors-org.s3.amazonaws.com/wp-content/uploads/2023/06/ORS-MSKI-RIG-ICM-Final.pdf), and the voting results from the 72 delegates (https://ors-org.s3.amazonaws.com/wp-content/uploads/2023/06/ORS-MSKI-ICM-Vote.pdf) are available on the ORS website. Eighteen Host Immunity

TABLE 1 Delegates of the Host Immunity Section, 2023 ORS MSKI ICM.

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Abbreviation: ORS MSKI ICM, Orthopaedic Research Society musculoskeletal infections International Consensus Meeting.

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& Rapid Fire questions surpassed a threshold consensus vote of 90%, and nine recommendations failed to achieve this threshold. Commentary on why consensus was not achieved on the nine questions and potential ways forward are described below.

3.1 | Question 1: Do joints have a natural microbiome, and does this affect joint infection?

One reason for the lack of consensus might be the inconsistent definition of the microbiome among different investigators, as this remains a controversial topic.[19] The ICM text correctly uses the term "microbiota" to refer to the viable bacteria, fungi, etc. However, the microbiome has also been defined more broadly in two ways: (1) as the entire habitat, including the community of bacteria, their genomes, and the surrounding metabolites, and so forth or as (2) the collection of genes and genomes from members of the bacteria (i.e., no viable bacteria). For example, it is likely that nucleic acids (rather than live bacteria) are found in the joint. ICM text refers to the microbiome as the collection of genes and genomes, not viable bacteria, as the authors tried to get this point across in the rationale, but it is possible that delegates did not fully appreciate the differences between these terms. By this definition, any site with free microbial DNA could be referred to as having a microbiome. Because the authors of this question view an active microbiota or microbiome community as a collection of viable organisms (commensals or pathogens), we voted that there is no natural joint microbiome. The ICM text nicely addresses the concerns regarding potential DNA contamination as the source of the microbiome, which has led to active controversies regarding a putative placental microbiome,[20,21] and so forth. If the microbiome was defined as a viable community of bacteria, the ultimate proof would be culturing of the community from the joint site, although this standard may be unduly restrictive as many bacterial species are not culturable with current techniques. Thus, research funding to answer this important question is warranted.

3.2 | Question 3: Does antibiotic therapy affect host immunity to MSKI?

As all ICM delegates have personal and professional experience with antibiotic therapy, it is likely that these biases influenced ICM delegate responses despite the lack of rigorous evidence and conflicting information regarding how antibiotic therapy affects host immunity, especially since effects were dependent on the antibiotic used. Delegates may also have pondered the indirect effects of antibiotic therapy on the gut microbiome, which can alter host immunity and may also compromise the ability to obtain repeatable outcomes. Every patient may have a different microbiome milieu, and the effects of the antibiotic therapy can vary between individuals, as well as its influences on the immune system.

TABLE 2 Tabled host immunity questions.

1. What tissues need to be studied to assess host immunity during MSKI?
2. Are patients susceptible and resistant to MSKI based on host immunity?
3. How does host immunity to MSKI change during treatment?
4. How does change in host immunity correlate with clinical outcome to MSKI?
5. What immunomodulatory factors in serum influence MSKI?
6. What is the best experimental approach to assess factors in serum?
7. Are there serum biomarkers associated with cure of MSKI?
8. What are the best models to study antibody responses during MSKI?
9. What populations of B cells need to be studied to evaluate MSKI and vaccine efficacy?
10. What is known about T-cell subsets in animal models of MSKI?
11. What are the best models to study phagocyte responses during MSKI?
12. How do macrophage populations within the synovium of a septic joint change during MSKI?
13. Do Trojan horse macrophages alter host immunity during MSKI?
14. Does the implant material alter host immunity to MSKI?
15. Does bone cement affect host immunity to MSKI?
16. Does implant fixation or loosening affect host immunity to MSKI?
17. What are the best models to study the effects of comorbidities on host immunity during MSKI?
18. Does allergy affect host immunity to MSKI?
19. Do other implants (e.g., pacemaker and stents) affect host immunity to MSKI?
20. Are there human-specific comorbidities that cannot be studies in animal models of MSKI?
21. Are there psychosocial impacts on host immunity to MSKI?
22. Does local infection management impact host immunity to MSKI?
23. Does revision surgery boost host immunity?
24. What is the role for humanized mice in assessing human immune responses to MSKI?
25. How does the antibody response to MSKI correlate with clinical outcome?
26. Does revision surgery boost host immunity?
27. How does host immunity to MSKI change during treatment in animal models? (also porcine vs small ruminants vs equine)?
28. Are there serum biomarkers associated with cure of MSKI?
29. What is the biomarker to predict efficient eradication of an MSKI before reimplantation in two-stage septic arthroplasty?
30. Are host immune responses associated with acute versus chronic MSKI?
31. Do nutritional markers affect host immunity in MSKI?

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32. Does the implant material alter host immunity to MSKI?
33. Does bone cement affect host immunity to MSKI?
34. Does implant fixation or loosening affect host immunity to MSKI?
35. Is there a correlation between *Staphylococcus aureus* nasal carriage and development of an *S. aureus*- MSKI?
36. Does local infection management impact host immunity to MSKI?
37. How does the antibody response to MSKI correlate with clinical outcome?
38. What tissues need to be studied to assess host immunity during MSKI?
39. How does change in host immunity correlate with clinical outcome to MSKI?
40. What are the best models to study responses by M0, M1, M2, and MDSC during MSKI?
41. What is the effect of biofilm on macrophage activation and antigen presentation?
42. How do macrophage populations within the synovium of a septic joint change during MSKI?
43. What is the best approach to study antibodies and B cell responses in animal models of MSKI?
44. What is known about T-cell subsets in animal models of MSKI?
45. What immunomodulatory factors in serum influence MSKI and what is the best methods to assess them?

Abbreviation: MSKI, musculoskeletal infections.

The following suggestions might be useful to reach a consensus on this question: (1) Antibiotic therapy studies in germ-free mice to control for the effect of the gut microbiome; (2) "omics" studies such as mass spectroscopy or RNA sequencing to understand the proteomic and transcriptional landscape altered by antibiotic therapies; (3) head-to-head comparisons between different antibiotic types to determine whether there is specificity in altered host immunity depending on the type of antibiotic used; and, (4) comparisons between patients with different microbiome milieus according to life-style, diet, health status, and gender.

3.3 | Question 5: Are there rigorous approaches for detecting and quantifying intracellular bacterial reservoirs?

With 87.5% agreement for the ICM recommendation that there are rigorous approaches for detecting and quantifying intracellular bacterial reservoirs, this question just missed the threshold for broad consensus. Given that these sophisticated intravital fluorescent and ultrastructural imaging experimental techniques only emerged in specialized labs recently, it is possible that some delegates are not fully aware of these novel studies. Delegates may also be aware that complementary molecular-based techniques are also in development that may help to overcome any ambiguities based on imaging alone. Thus, it is likely that future consensus will come from broad adoption of these state-of-the-art and developing technologies.

3.4 | Question 8: Does autoimmunity affect musculoskeletal infection in animal models?

Lack of agreement for the recommendation that autoimmunity does not affect MSKI in animal models largely came from 1/3 of the delegates who voted to abstain, suggesting that this topic is largely outside the scope of preclinical MSKI research, which is supported by the systematic review that only found two rigorous research articles on the subject: (1) mice with active collagen-induced arthritis that did not have exacerbated *S. aureus* PJI,[22] and (2) mice with hyperglycemic streptozotocin-induced type-1 diabetes that had a very modest increase in *S. aureus* implant-associated osteomyelitis.[23] Interestingly, this same study also found a much stronger increase in a type-2 diabetes model. As for the delegates who disagreed with the recommendation, this could have been due to bias coming from strong clinical data demonstrating that autoimmunity is a risk factor for MSKI. Nonetheless, a conclusion from the vote is that a few more rigorous studies are warranted to assess the effects of autoimmunity in animal models of MSKI are warranted.

3.5 | Question 10: Are there interventions beyond vaccination known to boost host immunity to MSKI in animal models?

The authors of the response to this question are somewhat surprised by the 10% disagreement with the recommendation that there is limited published data demonstrating that nutritional supplements, prebiotic and probiotic treatments, and moderate exercise have positive effects on infection reduction with various microbes in animal models. Thus, this group might be aware of non-peer-reviewed studies demonstrating otherwise. These dissenting opinions should be voiced, and more studies addressing this question should be published.

3.6 | Rapid fire questions 54–56, 58–63, & 65

As the lack of ICM consensus to the recommendations that the answers to these questions are "unknown" was due to a large delegate vote of "abstain" with very few "disagree" votes, we interpret this as a strong consensus affirming the recommendation that formal prospective research is needed to answer these questions.

3.7 | Question 57: Do bone cells mount an immune response during MSKI?

Intuitively it would seem that bone cells mount an innate immune response during MSKI. There have also been in vitro studies of

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TABLE 3 2023 ORS MSKI ICM host immunity questions.

Questions	Response/recommendation		Level of evidence	Delegate vote [%]		
				Agree	Disagree	Abstain
QUESTION 1: Do joints have a natural microbiome, and does this affect joint infection?	Yes	Microbial DNA has been detected in samples collected from joints of patients. The presence of microbial DNA in a region is no unexpected, however, since bacterial cell free DNA to be present in the systemic circulation. Whether or not live bacteria are present within the joint is unclear given the ease with which environmental contamination can influence measurement of samples with low bacterial load.	Low	46 [64]	18 [25]	8 [11]
QUESTION 2: Does gut microbiome affect host immunity during MSKI?	Yes	The microbiome is in constant contact with host immune cells at endothelial barriers. Interactions between immune cells and microbes or microbial proteins is known to cause immune responses that can be signaled around the body. Mouse models indicate the modifications to the gut microbiome can have a strong influence on osteomyelitis and periprosthetic infection.	Strong	70 [97]	0 [0]	2 [3]
QUESTION 3: Does antibiotic therapy affect host immunity to MSKI?	Unknown	There is limited and conflicted data to support a role for antibiotics to affect host immunity during implant infections. There was no evidence to support a role for antibiotics in systemic cytokine responses, and limited or conflicting evidence to support an effect of antibiotics on immune cell recruitment and implanted bone. There is evidence suggesting a negative impact of antibiotic therapy on the efficacy of immune responses by altering the gut microbiome. Although there is less evidence on the role of the microbiome on the incidence of implant infections. Given these results, further studies are warranted.	Low	53 [74]	13 [18]	6 [8]
QUESTION 4: Are there validated models of host immunity and with quantitative outcome measures?	Yes	There are validated models to study host immunity during orthopedic implant infections, with strong evidence for mouse models of prosthetic joint, tibial implant, osteomyelitis, and fracture fixation infections. However, there is only moderate evidence in rats, pigs, and rabbits for validation of orthopedic implant infection models. Frequent outcome measures in the validated animal models include measurements for cytokines production, immune cells populations, and antibody responses. However, our findings warrant further investigation to validate host immunity studies in medium-to-large animal models (e.g., rat, pigs, rabbits, and sheep) and their immunological outcomes.	Strong	65 [90]	2 [3]	5 [7]

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TABLE 3 (Continued)

Questions	Response/recommendation		Level of evidence	Delegate vote [%]		
				Agree	Disagree	Abstain
QUESTION 5: Are there optimal approach for detecting and quantifying intracellular bacterial reservoirs?	Yes	For in vitro experiments, current standards of detecting and quantifying intracellular bacterial include time laps dark field fluorescent microscopy of fluorescent mammalian cells (GFP, RFP, YFP, etc.) infected with a different fluorescent color bacteria, cultured with non-permeable antibiotic (i.e. gentamycin) to kill extracellular bacteria. These data should be validated with static outcomes including colony forming unit (CFU) assay, flow cytometry (i.e., Amnis ImagestreamTM imaging flow cytometry) for analysis of host cell-pathogen uptake in intracellular infections to provide bacterial counts together with statistical analyses to quantify bacterial load within mammalian cells. Transmission electron microscopy (TEM) utilized as a follow-up to flow cytometry is useful for correlating occupation of bacteria within mammalian cell cytosolic compartments. For in vivo experiments, multiphoton intravital microscopy of a fluorescent transgenic mouse (i.e., CMV-GFP) infected with compatible fluorescent bacteria (i.e., RFP, GFP) should be performed as the primary outcome with validation by stained histology (i.e., Gram stain) paraffin sections to define regions of interest (ROI) to be analyzed by TEM. For ex vivo assessment of clinical biopsies, histochemical staining again should be used to identify a region of interested to be interrogated by TEM. CFU assays and other methods to detect bacterial load (i.e., PCR) cannot be used to detect and quantify intracellular bacteria in tissue.	Strong	63 [87]	7 [10]	2 [3]
QUESTION 6: Can human immune responses be fully recapitulated in animal models of MSKI?	No	There is extensive literature that examines how the murine animal model has significantly contributed to our understanding of the immune responses and host–pathogen interactions during musculoskeletal infections (MSKI). However, the murine model has limitations in studying specific human immune responses. Mice engrafted with human immune cells, i.e., humanized mice are a better alternative. Recently, humanized mice have gained prominence as small animal surrogates to study human immune responses to infection. Indeed, MSKI studies with humanized mice are now being performed. But humanized mice also have limitations and cannot fully recapitulate human immunity. Large animal models (sheep, rabbit, horse, pig, and dog) are also routinely in the	Low	64 [89]	3 [4]	5 [7]

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TABLE 3 (Continued)

Questions	Response/recommendation		Level of evidence	Delegate vote [%]		
				Agree	Disagree	Abstain
		examination of MSKI and for testing orthopedic implants, biomaterials, and antibiotics. However, immune responses to infection in these animals are yet to be compared to human responses.				
QUESTION 7: Does anti-inflammatory medication influence MSKI in preclinical models?	Yes	The use of NSAIDs as an adjunct to antibiotics in mitigating the adverse effects of implant infections is supported by the evidence in rodent models with various infectious agents, including *S. aureus*. The influence of NSAIDs during implant infections included reduced bacterial burdens, decreased pro-inflammatory responses, and improved bone formation.	Strong	65 [90]	2 [3]	5 [7]
QUESTION 8: Does autoimmunity affect musculoskeletal infection in animal models?	Unknown	Limited evidence suggests that autoimmunity does not exacerbate MSKI in murine models.	Low	41 [57]	7 [10]	24 [33]
QUESTION 9: Is there an established immunization protocol (adjuvant, route of administration, time between boost) to assess novel vaccines for MSKI?	Yes	There is extensive literature on immunization protocols to assess novel vaccines in research animals. These protocols are specific for active immunization with: (1) biochemical (e.g., proteins, polysaccharides, peptides, oligosaccharides, or hapten-conjugate vaccines); (2) recombinant viral vectors, (3) DNA, and (4) mRNA lipid nanoparticles (LNP). There is also a very strong veterinary literature (peer-reviewed and commercial) for vaccination of domesticated animals, 1–3, which was not considered here.	Strong	59 [82]	4 [6]	9 [12]
QUESTION 10: Are there interventions beyond vaccination known to boost host immunity to MSKI in animal models?	Yes	There is peer-reviewed literature that largely demonstrates the efficacy of nutritional supplements, prebiotic and probiotic treatments, and moderate exercise on infection reduction with various microbes in animal models. However, the literature on animal models of MSKI is too limited to make a strong recommendation.	Low	55 [76]	7 [10]	10 [14]
QUESTION 11: Has the immune proteome been well-defined in animal models of MSKI?	No	Current literature examines the advancement in our understanding of the immune proteome in the setting of musculoskeletal infection (MSKI). Circulating antibodies represent a very important marker, which reflects the repertoire of antigens to which the immune system identifies as foreign (nonself). Antibodies are good biomarkers for diagnostic applications as they are stable to examine in experimental and clinical application. Essentially, immune proteomics is utilized for antigen discovery, which is effective in monitoring and diagnosing disease state, and even vaccine development.	Low	62 [86]	3 [4]	7 [10]

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TABLE 3 (Continued)

Questions	Response/recommendation		Level of evidence	Delegate vote [%]		
				Agree	Disagree	Abstain
QUESTION 12: Are there vaccines/passive immunizations for MSKI in animal models? (e.g. *S. aureus*)?	Yes	There is extensive peer-reviewed literature on vaccines and passive immunizations for MSKI, including *S. aureus* in experimental animals. There are also some veterinary commercial vaccines and passive immunizations for MSKI that are used in domesticated animals. Unfortunately, no such vaccines exist for humans.	Strong	64 [89]	3 [4]	5 [7]
QUESTION 13: Does prior MSKI shape immune responses to subsequent bacterial infection?	Yes	MSKI follows the theory of "original antigenic sin," which posits that prior exposure to pathogens shapes the immune response and affects the outcome of subsequent bacterial infection. The affect(s) can be positive (protective) or negative (pathogenic), and depends on the antigen(s), host genetics, and pathogenic circumstances.	Very Strong	70 [97]	0 [0]	2 [3]
QUESTION 14: How pathogen-specific is the antibody response to MSKIs?	Yes	The antibody-mediated immune response to MSKIs is highly pathogen-specific. There are several first-class studies characterizing the human and murine humoral response to *Staphylococcus aureus* (*S. aureus*)-related MSKIs. Such studies feature highly specific immune responses, which are dependent on many factors, including among other things: specific pathogen strain, site of infection, infection stage, host immune status, and comorbidities. While fewer studies have focused on pathogens outside of *S. aureus* in the setting of MSKI, there is strong evidence for the development of pathogen-specific antibodies during infection with a range of bacteria and fungi. Despite the specificity of host antibody-response to MSKI, such responses are rarely sufficient for microbial clearance, and hence might be of more value in the development of new diagnostic tools.	Strong	69 [96]	0 [0]	3 [4]
QUESTION 15: Are certain species more susceptible and resistant to MSKI based on host immunity than others?	Yes	Based on host immunity, we can assume that certain host species are more susceptible and resistant to MSKI than others. Specifically, larger animals such as pigs, sheep, and goats tend to replicate human disease more accurately than smaller animals such as rodents and rabbits. However, there is limited high-level evidence of this, and more studies must be performed.	Moderate	65 [90]	3 [4]	4 [6]
QUESTION 54: Are there any effective host immune responses against biofilm bacteria?	Unknown	-	Low	46 [66]	5 [7]	19 [27]

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TABLE 3 (Continued)

Questions	Response/recommendation		Level of evidence	Delegate vote [%]		
				Agree	Disagree	Abstain
QUESTION 55: How does host immunity against a pathogen change based on the location of the MSKI?	Unknown	-	Low	48 [68]	2 [3]	21 [29]
QUESTION 56: Are there any effective host immune responses against bacteria within the OLCN?	Unknown	-	Low	40 [56]	3 [5]	28 [39]
QUESTION 57: Do bone cells mount an immune response during MSKI?	Unknown	-	Low	42 [59]	8 [11]	21 [30]
QUESTION 58: What are the proven outcome measures of host immunity during MSKI?	Unknown	-	Low	44 [63]	2 [3]	24 [34]
QUESTION 59: Is there host immunity against small colony variants and persister cells?	Unknown	-	Low	45 [64]	3 [4]	22 [31]
QUESTION 60: What is known about antigen-specific T-cells in MSKI?	Unknown	-	Low	46 [65]	1 [1]	24 [34]
QUESTION 61: What are the important antigen-presenting cells in MSKI?	Unknown	-	Low	44 [65]	1 [1]	23 [34]
QUESTION 62: Is septic non-union associated with host immunity?	Unknown	-	Low	47 [66]	5 [7]	19 [27]
QUESTION 63: Do different pathogens influence host immunity against each other in polymicrobial MSKI?	Unknown	-	Low	50 [71]	4 [6]	16 [23]
QUESTION 64: Are monoclonal antibodies capable of eradicating biofilms in animal models of MSKI?	Unknown	-	Low	44 [62]	7 [10]	20 [28]
QUESTION 65: What are the best models to study T-cell responses during MSKI?	Unknown	-	Low	47 [66]	1 [1]	23 [32]

Abbreviation: ORS MSKI ICM, Orthopaedic Research Society musculoskeletal infections International Consensus Meeting.

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osteoblasts,[24] osteoclasts,[25] and osteocytes[26] demonstrating cytokine and chemokine production in response to bacteria and their byproducts. Thus, formal prospective studies in animal models of MSKI are warranted to fully characterize these immune responses and assess their functions.

3.8 | Question 64: Are monoclonal antibodies capable of eradicating biofilms in animal models of MSKI?

Although there have been several studies demonstrating the potential of anti-biofilm monoclonal antibodies (mAb),[27] experimental evidence proving that these mAb can eradicate biofilms in animal models remains to be formally demonstrated. This is partly due to the limited resolution of in vivo outcome measures to make such a bold and definitive conclusion. Thus, research funding is needed to enhance both the sensitivity and specificity of biofilm eradication measurements and the evaluation of promising mAb in validated animal models of MSKI with quantitative outcome measures of their biofilm infection.

4 | DISCUSSION

The "Host Immunity" Section of the 2023 ORS MSKI ICM (Table 1) presented 15 questions with recommendations and level of evidence and another 12 questions to be voted on with a recommendation of "Unknown" without systematic reviews to 72 international experts using the Delphi method to reach consensus on. However, despite its major strengths in rigor, scope, and inclusiveness, it is recognized that the Delphi method has some inherent weaknesses. The greatest among these weaknesses was the bias in which 45 important host immunity questions (Table 2) were tabled by the ICM due to a lack of human resources for performing additional systematic reviews. The tabling of these questions should not be interpreted to mean that they are less significant than the host immunity questions that were presented to the ICM for voting, and it is our recommendation that future ICMs complete the Delphi method on these questions.

Another weakness was that the ICM design did not allow for anonymous discussion, which could have impeded free expression by ICM delegates in other Sections who are less knowledgeable about host immunity. It is likely that the bulk of "abstain" votes came from these delegates, which resulted in 14 Host Immunity questions failing to achieve the ≥90% agreement threshold of ICM consensus. It is also noteworthy that some scientific areas are advancing very rapidly, such as imaging and quantification of intracellular bacteria and host immunity against biofilm. Thus, a degree of scientific uncertainty and unknowns about recently published and unpublished data could be responsible for a lack of consensus on these questions.

Inherent to the Delphi method, participants considering questions outside their area of expertise could reach an incorrect consensus with a high level of confidence based on a lack of knowledge. The participants in the Host Immunity Section included the world leaders in all areas of microbial pathogenesis and immunology that were covered by the 27 questions posed for ICM voting. Thus, it is unlikely that the consensus reached was based on a lack of knowledge.

Lastly, the Delphi method is best for addressing single scalar topics. When complex interdependent areas of knowledge have incongruous results, such as experimental data from in vitro experiments, animal models and clinical studies, the possibility exists that a consensus is impossible. The "Host Immunity" Section was very deliberate in refining the wording of the questions and responses to avoid these pitfalls.

As the state of host immunity knowledge expands and related strategies are adopted by the MSKI field, the "Host Immunity" Section expects new questions to arise, and the responses to the current questions will advance justifying clinical recommendations in the future. We anticipate another ICM in the future to address the tabled questions and update the consensuses from the 2023 ORS MSKI ICM. Alternatively, given the technical and logistical challenges in completing an ICM, MSKI stakeholders could hold meetings similar to the Implant Wear Workshops that were held in 2000,[28] 2007,[29] and 2013[30] that resulted in very high impact text books, review articles, and subsequent primary research papers, which many believe have addressed and helped mitigate the clinical problems of periprosthetic osteolysis and aseptic loosening.

5 | CONCLUSION

It is anticipated that the data, rationale, and response for each question will provide researchers with a higher level of understanding of host immunity to MSKI, which can lead to a sharper focus on the greatest research needs in this field. Regrettably, there are more important questions that could be answered by this initial ICM on MSKI host immunity, and subsequent work is needed to address the tabled questions. Studies dedicated to advancing our understanding of host immunity and its role in MSKI are urgently required for better diagnostics, prognostics, prophylaxis, and therapies. It is also noted that biomedical research funding agencies and the pharmaceutical industry should recognize these areas as priorities.

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REFERENCES

1. Kurtz SM, Lau E, Schmier J, Ong KL, Zhao K, Parvizi J. Infection burden for hip and knee arthroplasty in the United States. *J Arthroplasty*. 2008;23:984-991.
2. Stulberg JJ. Adherence to surgical care improvement project measures and the association with postoperative infections. *JAMA*. 2010;303:2479-2485.
3. Schwarz EM, Parvizi J, Gehrke T. The 2018 International Consensus Meeting on musculoskeletal infection: research priorities from the general assembly questions. *J Ortho Res*. 2019. In Press.
4. Cram P, Lu X, Kates SL, Singh JA, Li Y, Wolf BR. Total knee arthroplasty volume, utilization, and outcomes among Medicare beneficiaries, 1991-2010. *JAMA*. 2012;308:1227-1236.
5. Rosas S, Ong AC, Buller LT, et al. Season of the year influences infection rates following total hip arthroplasty. *World J Orthop*. 2017;8:895-901.
6. Azzam K, McHale K, Austin M, Purtill JJ, Parvizi J. Outcome of a second two-stage reimplantation for periprosthetic knee infection. *Clin Orthop Rel Res*. 2009;467:1706-1714.
7. Patel R. Periprosthetic joint infection. *N Engl J Med*. 2023;388: 251-262.
8. Bagnoli F, Bertholet S, Grandi G. Inferring reasons for the failure of *Staphylococcus aureus* vaccines in clinical trials. *Front Cell Infect Microbiol*. 2012;2:16.
9. Proctor RA. Is there a future for a *Staphylococcus aureus* vaccine? *Vaccine*. 2012;30:2921-2927.
10. Muthukrishnan G, Wallimann A, Rangel-Moreno J, et al. Humanized mice exhibit exacerbated abscess formation and osteolysis during the establishment of implant-associated *Staphylococcus aureus* osteomyelitis. *Front Immunol*. 2021;12:651515.
11. Schwarz EM, Alt V, Kates SL. The 1st International Consensus Meeting on periprosthetic joint infection. *J Orthop Res*. 2014; 32(suppl 1):S1.
12. Cats-Baril W, Gehrke T, Huff K, Kendoff D, Maltenfort M, Parvizi J. International consensus on periprosthetic joint infection: description of the consensus process. *Clin Orthop Relat Res*. 2013;471:4065-4075.
13. Parvizi J, Tan TL, Goswami K, et al. The 2018 definition of periprosthetic hip and knee infection: an evidence-based and validated criteria. *J Arthroplasty*. 2018;33:1309-1314.
14. Schwarz EM, Parvizi J, Gehrke T, et al. 2018 International Consensus Meetinginternational consensus meeting on musculoskeletal infection: research priorities from the general assembly questions. *J Orthop Res*. 2019;37:997-1006.
15. Saeed K, McLaren AC, Schwarz EM, et al. 2018 international consensus meeting on musculoskeletal infection: summary from the biofilm workgroup and consensus on biofilm related musculoskeletal infections. *J Orthop Res*. 2019;37:1007-1017.
16. Schwarz EM, McLaren AC, Sculco TP, et al. Adjuvant antibiotic-loaded bone cement: concerns with current use and research to make it work. *J Orthop Res*. 2021;39:227-239.
17. Fowler VG, Allen KB, Moreira ED, et al. Effect of an investigational vaccine for preventing *Staphylococcus aureus* infections after cardiothoracic surgery: a randomized trial. *JAMA*. 2013;309: 1368-1378.
18. Nishitani K, Ishikawa M, Morita Y, et al. IsdB antibody-mediated sepsis following *S. aureus* surgical site infection. *JCI Insight*. 2020;5(9):e141164.
19. Marchesi JR, Ravel J. The vocabulary of microbiome research: a proposal. *Microbiome*. 2015;3:31.
20. Panzer JJ, Romero R, Greenberg JM, et al. Is there a placental microbiota? A critical review and re-analysis of published placental microbiota datasets. *BMC Microbiol*. 2023;23:76.
21. Kennedy KM, de Goffau MC, Perez-Muñoz ME, et al. Questioning the fetal microbiome illustrates pitfalls of low-biomass microbial studies. *Nature*. 2023;613:639-649.
22. Trikha R, Greig D, Sekimura T, et al. Active rheumatoid arthritis in a mouse model is not an independent risk factor for periprosthetic joint infection. *PLoS ONE*. 2021;16:e0250910.
23. Farnsworth CW, Shehatou CT, Maynard R, et al. A humoral immune defect distinguishes the response to *Staphylococcus aureus* infections in mice with obesity and type 2 diabetes from that in mice with type 1 diabetes. *Infect Immun*. 2015;83:2264-2274.
24. Marriott I. Osteoblast responses to bacterial pathogens: a previously unappreciated role for bone-forming cells in host defense and disease progression. *Immunol Res*. 2004;30:291-308.

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25. Madel MB, Ibáñez L, Wakkach A, et al. Immune function and diversity of osteoclasts in normal and pathological conditions. *Front Immunol*. 2019;10:1408.
26. Yang D, Wijenayaka AR, Solomon LB, et al. Novel insights into *Staphylococcus aureus* deep bone infections: the involvement of osteocytes. *mBio*. 2018;9(2):e00415-18.
27. Raafat D, Otto M, Reppschläger K, Iqbal J, Holtfreter S. Fighting *Staphylococcus aureus* biofilms with monoclonal antibodies. *Trends Microbiol*. 2019;27:303-322.
28. Wright TM, Goodman SB, eds. Implant wear in total joint replacement: clinical and biologic issues. *Material and Design Considerations*. 1st ed. American Academy of Orthopaedic Surgeons; 2001.
29. Goodman SB, Wright T. 2007 AAOS/NIH osteolysis and implant wear: biological, biomedical engineering, and surgical principles. introduction. *J Am Acad Orthop Surg*. 2008;16(suppl 1):x-xi.
30. Goodman SB, Wright TM. Editorial comment: ABJS Carl T. Brighton workshop on implant wear and tribocorrosion of total joint replacements. *Clin Orthop Relat Res*. 2014;472: 3650-3651.

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